

Remote Sensing Thermometer

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Ever since the invention of thermometer, various techniques have been developed and used to measure temperature of solid, liquid and gaseous matters. But none of these techniques could measure the temperature from a remote place, which sometimes becomes a necessity, particularly when the object under test is in a dangerous or inaccessible area. Presented here is a remote sensing thermometer to measure the temperature from a remote place.

Circuit and working

Fig. 1 shows block diagram of the remote sensing thermometer. The temperature of the object under test is sensed by a temperature sensor. The resultant voltage developed at the output of the sensor cannot be sent to a remote destination through normal wired path as its magnitude is generally very low and would be highly attenuated during transit.

One way to solve this problem is to convert the sensed voltage into equivalent frequency by using a voltage-to-frequency (V-F) converter and send the same to the remote end through a transmitter. At the remote end, a frequency-to-voltage (F-V) converter is used to retrieve the original signal from the received frequency-encoded signal for display or control process.

Fig. 2 shows the circuit of the transmitter unit. It comprises temperature sensor LM35 (IC1), two CA3140 operational amplifiers (IC2 and IC3), voltage-to-frequency converter KA331 (IC4), opto-coupler MC2TE (IC5), 433MHz ASK transmitter, regulators 7805 (IC6) and 7905 (IC7), and a few discrete components.

Temperature sensor IC1 develops

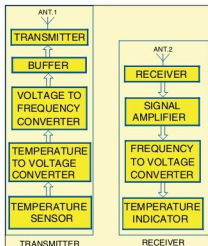


Fig. 1: Block diagram of the remote sensing thermometer

Test Points

Test point	Details
TP0, TP6	0V, GND
TP1, TP7	5V
TP2	-5V
TP3	Voltage corresponding to temperature (refer the datasheet for exact values)
TP4	Pulse train with frequency corresponding to input temperature
TP5	Input pulse train to the transmitter module
TP8	Received signal
TP9	Sensed voltage

a voltage at its output pin 2, which is equivalent to the temperature being sensed. It can measure from -55°C to 150°C. LED1 connected to pin 3 of IC1 raises its output by a few hundred millivolts. Thus the voltage obtained at the output is the sum of LED voltage and the sensed voltage. This voltage enhancement is required to transmit the frequency-encoded signal for 0°C and below-0°C temperatures.

The output from IC1 is fed to input of the unity-gain inverting amplifier

